

		Option D: $\left[\frac{1}{2eVv^2}\right] = \frac{1}{\text{kg m}^2 \text{s}^{-2} \text{ m}^2 \text{s}^{-2}} = \text{m}^{-2} \text{s}^2$
20	A	The wire carries a constant electric current, I .

No.	Answer	Solution
		$I = Anvq$ $= \frac{\pi}{4} d^2 nvq \quad \left(\because A = \pi r^2 = \pi \left(\frac{d}{2} \right)^2 = \frac{\pi}{4} d^2 \right)$ $\therefore v \propto \frac{1}{d^2}$ $\text{i.e. } v = \frac{K}{d^2}$
31	D	Resistance of voltmeter is equal to resistance of R ₁ + R ₂